OPTIONS-BUYING SCOPE (D4) + PER-SYMBOL MULTI-STYLE CROSSCHECK — real-evidence review

Session 27 · 2026-10-27 · owner scope change + evidence crosscheck (web-researched) Owner decision: no cash equities, no futures on stocks — **options only, BUYING only**, on NIFTY, SENSEX and the stock roster. Each symbol runs the compatible styles via its own cells; online mode picks what works now and exits immediately on adverse news. **Covers:** D4 roster re-baseline · 4-of-8 style physics · per-symbol architecture check · NSE/SEBI rulebook constraints · **the "does it work / how profitable" evidence search** · regime router · exit ladder 1–6 · cell-count math. Every claim classified: measured / web-verified / unverified.

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§1 · What the owner decided (D4) and what it changes

Owner said	Effect
"not using equities & futures on stocks"	**D1 re-opened** — 57 NSE cash-equity cells and stock-futures cells drop
"only options, buying only"	stock cells become **options-buying cells** on optionable names
NIFTY + SENSEX stay	index options (NFO / BFO) — compatible with buying (Session 23, D3)
"each symbol has its own agent with all these styles"	per-symbol × **4 compatible styles** = specialist cells (Session 22 preserved)
"online figures out which works now and does those"	deterministic **regime router** (§6)
"mid-trade news → immediately close"	**exit ladder 1–6** (§7)
FX / commodities / crypto	unchanged — 14 cells (7 FX + 4 commodity + 3 crypto)

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§2 · Buy-only vs the 8 options styles — only 4 fit (option physics)

The core physics: buying options PAYS theta every day. You cannot harvest what you are paying.

#	Style	Fits buy-only?	Why
1	**delta-direction** (long calls/puts)	✅ **YES**	pure directional buying
2	**vol-alpha (vega)**	✅ **YES**	buy cheap vol (long vega) — buy-only version works
3	**gamma-scalper**	✅ **YES**	buy straddle → scalp delta — long gamma is buying
4	**hedge-executor**	✅ **YES**	buying protective puts = exactly this
5	theta-harvest	❌ NO	it means **selling** premium — a long book *pays* theta
6	basis/calendar	❌ NO	a calendar = long far + **short near** — contains a short leg
7	expiry/pin	❌ NO	pin-harvest = selling into expiry (theta capture)
8	options-MM	❌ NO	quoting = selling + already deferred (Session 25: MM impossible on our venues)

Options roster after D4: 4 of 8 — delta-direction · vol-alpha · gamma-scalper · hedge-executor. The other 4 are excluded by the owner's own buy-only rule (option physics, not architecture).

Honest economics of buy-only: every day held, theta bleeds the premium. Long-only wins need direction or vol move > decay before expiry. Loss is capped at premium paid — the governor's worst case is **known at entry** — but the win-rate bar is higher than selling. Capped-loss / positive-omega is a legitimate choice; it is what fits the risk rule.

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§3 · Per-symbol architecture crosscheck — "one agent per symbol with all these styles"

Correction first (Session 22 evidence): not one agent holding all styles simultaneously — **one specialist cell per (symbol × style)**, because ScaleZero-style negative interference kills multi-style simultaneous weights. What the owner describes = 4 cells per symbol, with the regime router enabling/disabling them.

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SYMBOL (e.g. NIFTY)
 ├─ cell NIFTY-delta-direction   ─┐
 ├─ cell NIFTY-vol-alpha          ├─ all offline-researched (Mode R, 5 methods)
 ├─ cell NIFTY-gamma-scalper      │   promoted through E4 → frozen artifacts (Mode T)
 └─ cell NIFTY-hedge-executor    ─┘
        ▲
        └── REGIME ROUTER (deterministic rules, default = all off) enables 0 or 1 at a time
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Question	Verdict	Evidence class
One specialist cell per symbol × style?	✅ yes — Session 22 evidence (negative interference)	measured (internal)
All 4 styles researched offline per symbol?	✅ yes — Mode R cell, 5 methods, frozen artifacts	design (Session 26)
Online picks the working style?	✅ deterministic regime router, default off	design (§6)
Does multi-style breadth help?	✅ **web-verified** — CTA multi-strategy literature, §5.5	web-verified
Will it be profitable?	⚠️ **unknown until backtest + shadow** — see §8	unproven

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§4 · The instruments — what NSE/BSE actually allow (verified 2026-08)

Fact	Source	Consequence for us
F&O on **6 indices + more than 200 securities**, only those meeting SEBI eligibility	NSE (updated 11-Aug-2026)	**not all 57 names have options** — screen at onboarding
One weekly index expiry per exchange (NIFTY only on NSE; SENSEX on BSE)	SEBI reform	fewer short-dated contracts; stock options = monthly
Option premiums paid fully upfront	SEBI	buy-only = the cheaper side (sellers post margin)
Lot sizes sized to **₹15–20 lakh contract value** (NIFTY = 65 from Jan 2026)	NSE FAOP70616	position-sizing math changes; capital per trade is large
Ban period: MWPL > 95% → reduce-only, no new positions	SEBI May-2025 circular	**hard entry gate: check ban status before every entry**
Intraday position monitoring ≥ 4 snapshots/day	SEBI Apr-2025	E5 checks limits intraday, not just EOD

Onboarding screen required: intersect the 57 names with the option master + **liquidity filter** (stock options in India are far thinner than index options — wide spreads, sometimes no bid). Honest expectation: **tradeable set < 57**. Also new: **O3 gets sharper** — a NIFTY cell is one underlying, but its tradable instruments are thousands of strike×expiry contracts; the router picks the strike, not the cell.

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§5 · THE EVIDENCE CROSSCHECK — "does it work, how profitable, real web evidence"

5.1 The retail base rate (SEBI's own study — the world we are NOT joining)

Finding	Source
93% of 1.13 crore individual traders lost ₹1.81 lakh crore in F&O over FY22–FY24 (avg ~₹2 lakh/person)	SEBI/PwC study, Sep-2024
91.1% lost in FY24 alone (~73 lakh traders, avg net loss ₹1.20 lakh incl. costs)	same
>80% of options losses came from 17% "high-value" traders; **93.5% of them lost**	same
Only ~1% of traders earned >₹1 lakh	SEBI PR 22/2024

Read correctly: this is the base rate for *unsystematic* retail — discretionary entries, no pre-defined exits, leverage, no cost accounting. It is the population our deterministic governor, exit ladder, and E4 gates exist to avoid. It is **not** evidence that systematic buy-only cannot work — and it is **not** evidence that it can.

5.2 Strategy-level evidence on NIFTY options (both sides, net of costs)

Study	What it tested	Result
SSRN Pillai 2026 — VRP on Nifty 50, realistic frictions	**4 short-vol strategies, 119 expiries, 2015–2025**	❌ **ALL negative net of costs** — straddles ≈ −44 to −45%/yr; put-write −0.9%/yr **with a 91.6% win rate**
Nifty short-straddle retail backtest	systematic short straddle	⚠️ 68% win, **+14.8% pre-tax** — costs and tail risk decide the sign
IRE decade study (2015–2025)	weekly index behaviour, Fri→Thu	avg weekly move only **+0.12%** (no free directional edge); moves >±300 pts occur **28.7%** of weeks (risk is fat)
Angled short straddle (Indian market, 2021)	selling variant vs plain short straddle	+18 ppt success, +57% profitability — **still a selling strategy**

The brutal honest reading: after costs, **the documented NIFTY strategies lose on BOTH sides** (Pillai: sellers all negative; SEBI: buyers 91% negative; retail short-straddle backtests flip sign once frictions are honest). **There is no free side. Edge must come from selection, timing, and exits — not from which side of the trade you take.**

5.3 Evidence on the BUY side specifically (what we chose)

Evidence	Result	Class
IV-crush studies on long straddles	buying vol into events, holding through the print → **systematically pays for the implied>realised gap**	web-verified
LEAP backtest (Ryan Tolone, 2024)	naive long-dated deep-ITM calls **can underperform simply holding stock**; "wins concentrated in deep-OTM cells — exactly the cells with worst path-dependence"	web-verified

Evidence	Result	Class
Deep-OTM monthly put buying, BTC/ETH 2018–2026	tail insurance: **BTC +97.6% but ETH −73.1%**, very low win rates, top-5 trades >10× net profit → "not a stable source of profit"	web-verified
Monash VRP study (S&P 500)	systematic **long**-option books exist but the VRP literature's harvest sits with **sellers**; top systematic strategies are short-side	web-verified
Long-call retail backtest sites (QQQ/SPY, 85%/71% win)	⚠️ **not credible** — no honest cost model, CAGR figures in the hundreds of % — discard	unverified
Indian equities long-only momentum, net of costs (QuantifiedTrader)	systematic long-only equity momentum survives costs in India — but that is **stock**, not options	web-verified

Buy-only verdict: every credible independent study of *mindless* long-option buying loses (theta + IV crush). The buy side only works with **(a) cheap vol entry, (b) catalyst/timing, (c) strict exit before decay outruns edge** — exactly what the 4 buy-compatible styles and exit ladder are for. That is a design hypothesis until Mode R proves it on our data.

5.4 What the owner's system adds that the losing population does not have

Documented failure (evidence above)	Our counter-measure	Class
Discretionary entries, no rules	deterministic cells, frozen artifacts, E5 governor	design
No exit discipline (hold losers)	exit ladder 1–6, pre-flatten before prints	design
Unknown worst case	**loss capped at premium — known at entry**	physics of buy-only
Over-trading thin contracts	liquidity + optionability screen at onboarding	design
Ban-period / limit violations	ban check in entry gate, intraday MWPL monitoring (E5)	design
Strategy chosen by mood	regime router, default = all off	design
Unaccounted costs (STT, slippage, spread)	E7 cost model in the **label** before any promotion	design

5.5 Does the multi-style, multi-symbol architecture itself have evidence? — YES

Study	Finding
CME/Efficient Capital CTA diversification (2014)	diversification **continuously adds value**; optimal = 2–5 managers per style; complementary-style portfolios beat concentrated ones
Campbell White Paper (Feb 2015)	clear diversification benefit in strategy portfolios; cross-strategy correlation **weakens when momentum opportunities exist**
CFA Digest (2018) — CTAs: superior performance or diversification only	median pairwise CTA correlation **0.08**; adding managed futures raised return 8.76% → up to **11%** while cutting vol 9.63% → **8.66%**
Campbell multi-strategy flagship	**Sharpe > 1** partly from netting/diversification across 4 strategies; 144 models with a diversification criterion for admission
Grinold–Kahn breadth law (Session 26)	**IR ≈ IC × TC × √BR** — 59 symbols × 4 styles raises independent-bet breadth; same IC → materially higher IR

Architecture verdict: the multi-cell, multi-style breadth design is the part with the strongest external evidence. The buy-only options edge itself is the unproven part.

5.6 Honest bottom line on "how profitable"

Question	Answer
Does the architecture work?	✅ **supported** — specialist cells (measured) + multi-strategy diversification (web-verified, §5.5)
Do long options make money by default?	❌ **NO** — SEBI 91–93% lose; IV crush; Pillai shows even the documented NIFTY strategies lose net of costs

Question	Answer
Is there a credible profit number today?	❌ **No — anyone quoting one pre-backtest is guessing.**
What is honestly known?	worst case = premium paid (capped) · base rate = ~91–93% of traders lose (they are unsystematic) · breadth/diversification raises IR · our gates only promote what survives CPCV + DSR ≥ 1 + shadow
What would make it profitable?	Mode R must find, **on our own data**, entries with expected value > (premium decay + STT + spread + slippage) — then E4 promotion, then shadow proves it live
Failure condition (defined up front)	if the research cell cannot clear DSR ≥ 1 on buy-only option strategies after a full search budget → **the scope does not trade**; that is the gate working, not a bug

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§6 · Online mode — the deterministic regime router ("figures out what works now")

Regime state (computed, rules-only)	Enabled style
trend + normal vol	**delta-direction**
vol percentile low + event within N days	**vol-alpha** (buy before catalyst)
high realized vol, intraday	**gamma-scalper**
portfolio DD rising / event risk	**hedge-executor** (buy protection)
ambiguous / news window / ban period / no liquidity	**ALL OFF — no entries**

#	Rule
R1	regime→style map is itself proven offline in Mode R (DSR-gated like any candidate)
R2	router = **rules, zero ML** — deterministic governor doctrine (E5)
R3	cells stay one-style specialists — router only enables/disables, never re-weights (Session 22)
R4	**default = everything off** — not trading is a valid position

Honest risk: style-timing rules overfit easily → R1 exists for that reason.

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§7 · Exit ladder 1–6 ("mid-trade news → immediately close")

#	Exit	Fires when
1	**Pre-flatten / never-enter**	print due within ±5 min → no entries; existing positions closed **before** the window
2	**Hard stop-loss**	premium −30–50% (config per style) → close immediately
3	**Time-stop**	DTE < X, or decay faster than edge → close (theta works against you daily)
4	**News-flatten (mid-trade)**	adverse event state fires → close now — owner's rule, event trigger in E5
5	**Liquidity gate on exit**	exit only when a real bid exists; if none → degrade to underlying hedge (closing, not opening)
6	**Kill-switch**	E5 ladder breach → all cells off

Caveat: closing *during* the news moment pays the widest spread of the day — rule 1 (flatten **before** the print) matters more than rule 4. With buy-only, "close immediately" saves the **remaining premium**; max loss was always capped at what was paid.

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§8 · Cell-count math after D4 (honest ranges)

Group	Formula	Count
Index options	2 (NIFTY, SENSEX) × 4 styles	**8**
Stock options	**N × 4**, N = optionable ∩ liquid subset of the 57	**8 ≤ N ≤ 57 → 32…228** (N known only after onboarding screen)
FX + commodities + crypto (unchanged)	7 + 4 + 3	**14**
Total	22 + N×4	**54 … 250**

⚠️ **O7 surfaced:** hedge-executor (buying protection on portfolio drawdown) is arguably **one portfolio-level cell**, not 57 per-symbol copies. If so: (2+N)×3 + 1 + 14. Flagged for owner — affects every count above.

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§9 · Provenance

Claim	Class
SEBI 93% / 91.1% / ₹1.81 lakh cr / 17% high-value losses	**Web-verified** — SEBI-PwC Sep-2024, PR 22/2024
Pillai 2026: 4 short-vol strategies all negative net of costs, 119 expiries	**Web-verified** — SSRN
LEAP underperformance, deep-OTM tail-insurance instability, IV-crush losses	**Web-verified** — Tolone 2024, HTX 2026, IV-crush literature
Long-call backtest sites (85% win, CAGR 300%+)	**Unverified — discarded** (no credible cost model)
CTA/multi-strategy diversification (corr 0.08, 8.76→11%, Sharpe>1)	**Web-verified** — CME 2014, Campbell 2015, CFA Digest 2018
NSE 6 indices + >200 securities; single weekly expiry; upfront premium; MWPL ban; NIFTY lot 65	**Web-verified** — NSE, SEBI circulars (Oct 2024–May 2025)
4-of-8 style compatibility	**Derived from option physics** — not an empirical study
Per-symbol specialist cells (vs one generalist)	**Measured (internal)** — Session 22 ScaleZero evidence
Regime router, exit ladder, gates	**Design requirements** — not yet measured
Profitability of this scope	**UNPROVEN until Mode R backtest + E4 + shadow** — §5.6 failure condition defined

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§10 · Open decisions this session creates

- **D4** — cash equities dropped → options-buying roster (this artifact is the working spec).
- **O6** — add "scalping" to the FX roster (30 → 31 locked styles)?
- **O7** — hedge-executor: per-symbol cell × N, or **one portfolio-level cell**?
- Carry-forward: O1 (predictions 0 cells), O3 (index F&O granularity — sharpened by §4), O4 (7 off-index names), O5 (styles per instrument — D4 now implies **4 per optionable symbol**).
- **Onboarding screen** — intersect 57 names with option master + liquidity filter → sets N.

Artifact of Session 27. Companion to QUANT-TRADING-ALPHA-EXPLAINED.md (Session 26), FX-SCALPING-BUILD-PLAN.md (Session 25), FIRST-BATCH-NON-LLM-AGENTS.md (needs D4 rewrite).